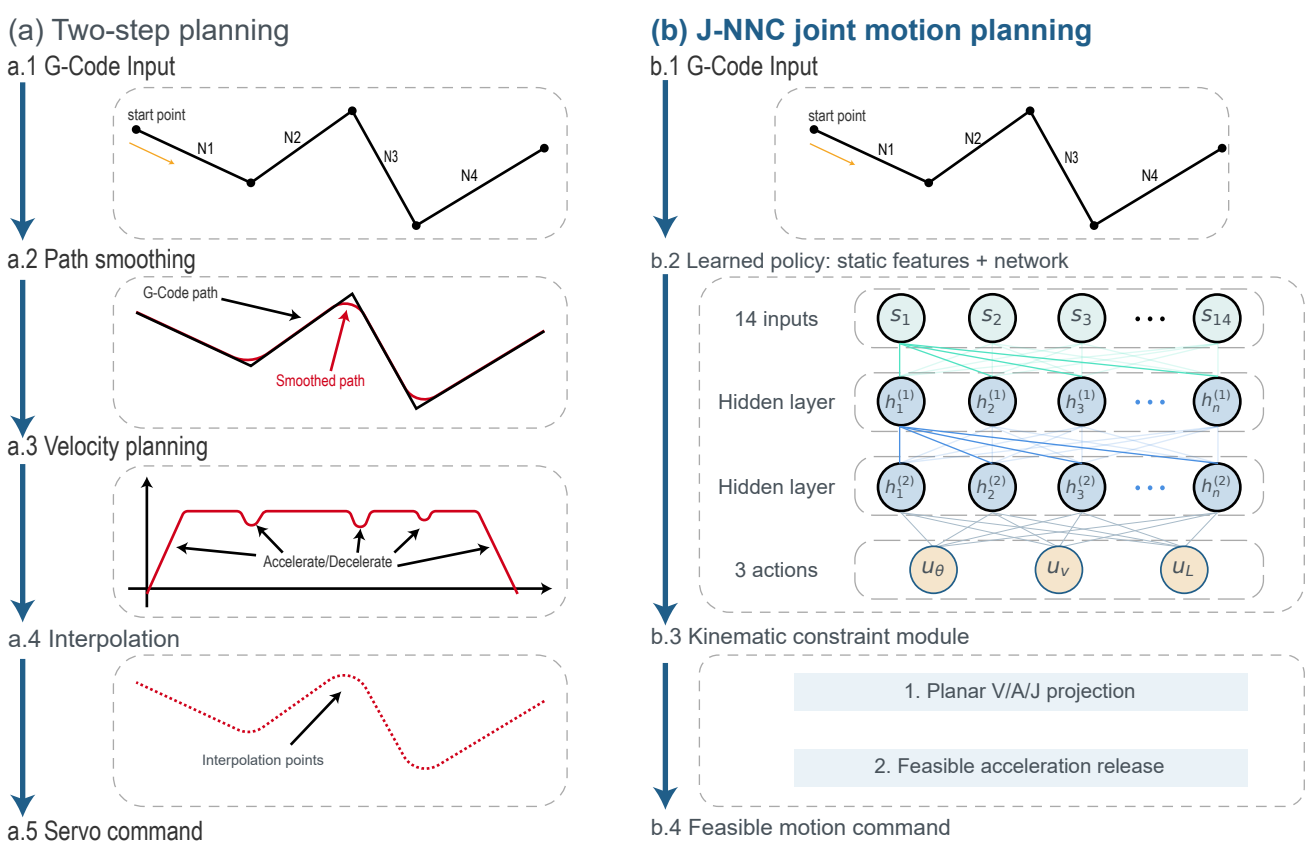

(a) Two-step planning
a.1 G-Code Input
start point
N1
N2
N3
N4
a.2 Path smoothing
G-Code path
Smoothed path
a.3 Velocity planning
Accelerate/Decelerate
a.4 Interpolation
Interpolation points
a.5 Servo command
(b) J-NNC joint motion planning
b.1 G-Code Input
start point
N1
N2
N3
N4
b.2 Learned policy: static features + network
14 inputs
Hidden layer
Hidden layer
3 actions
b.3 Kinematic constraint module
1. Planar V/A/J projection
2. Feasible acceleration release
b.4 Feasible motion command